AceEngineer

Seven problems every small operator has, and what an engineer can actually do about each one.

This list did not come from a market study. It came from reading what operators running shallow, marginal, conventional wells wrote about their own fields this summer — the cards they pulled, the bearings they lost, the parts they could not find. **Every problem below is quoted, not imagined.**

WHERE THIS CAME FROM

We work rod-lift and production engineering as software: physics that runs on your own well data, not a dashboard subscription. Last month one of our analyses was posted publicly and corrected within hours by a rod-lift specialist who teaches a dyno class. He was right. We fixed it, and we changed how we work — nothing goes out now without the evidence attached to it.

So this is a field note, not a pitch deck. Where we can do something, it says so plainly. Where we cannot, it says that too.

WHAT WE HEARD

SO-LIFT-001

ARTIFICIAL LIFT

The well pumps off in five minutes and there is no controller on it

> "*What happened to my pump card over the course of this 5 minutes?*"
>
> Gulf Coast operator, on a unit with no pump-off controller

THE GAP

Good advice came back fast: slow the unit, or set a shutdown below 65–75% fillage. Both assume a controller the well does not have — and neither is tuned to *this* well. The setpoint that is right for your pump, your rod string and your inflow is a calculation, not a rule of thumb.

WHAT WE DO

Run your card sequence through a full surface-to-downhole conversion, extract fillage stroke by stroke, and return a specific shutdown setpoint and slow-down schedule — with the production you give up and the pump life you buy, stated separately so you can decide.

SO-LIFT-002

ARTIFICIAL LIFT

Is that noise on the card real, or is it the instrument?

> *"Could the noise in this corner be from gas in the pump decompressing… or a delay in valves closing? There isn't really any noise here when the pump is full."*
>
> Operator, reading a derived downhole card

THE GAP

On an accelerometer-based dyno, position is integrated twice from acceleration, so error accumulates across a stroke — and the corner right after the stroke-to-stroke discontinuity is exactly where the wave equation has not settled. Some of what looks like a pump tag is arithmetic. Some of it is a pump tag.

WHAT WE DO

Put an error budget on the derived card and mark which segments carry enough numerical confidence to diagnose on. You stop chasing artefacts and you stop dismissing real signals.

SO-MAINT-002

MAINTENANCE

The low-volume wells get no attention until a neighbour calls

> *"This low volume well is generally at the bottom of the attention list due to its low IP rate, low accumulated recovery, and low production… understanding each well as an individual takes time and time is on no one's side."*
>
> Illinois Basin operator, ~150 wellbores

THE GAP

This is not an equipment problem. One person cannot hold 150 wells in their head, so the marginal ones are invisible until they fail loudly. What is missing is a ranking.

WHAT WE DO

Score every well in the field on health and fillage and return one ordered list: *drive to these five today*. It is the single highest-leverage thing on this page, and it needs no new hardware.

SO-MAINT-001

MAINTENANCE

The walking beam left and took the saddle bearing with it

> *"The walking beam went for a walk about and took the saddle bearing with it."*
>
> Illinois Basin operator

WHAT WE DO

Torque and gearbox loading checks against the unit's actual rating, so you find out which of your units is running outside its envelope before the structure tells you. We will ask for the gearbox designation rather than infer it from the engine — we have made that mistake in public once and do not intend to repeat it.

SO-SUPPLY-001

SUPPLY CHAIN

One missing seat nipple idles a rig day

> *"Guide wires, deadman, rods, tubing, rod subs, tubing subs, mud anchor, insert pump, seat nipple, rod boxes, tubing collars, gas anchor, barrel — 'word salad.'"*
>
> Illinois Basin operator, describing a workover

THE GAP

That list is a bill of materials, and every operator in the basin holds their own partial copy of it. Dead stock on one lease, a stopped job on the next. See the proposal below.

SO-CAPITAL-001

CAPITAL

"Workovers will add 30 BOPD" — will they?

> *"Buying out an existing mortgage held by the present contract operating company. They are falling short and failing at their duty of maintaining and increasing production."*
>
> Operator seeking growth capital against ~150 wellbores

WHAT WE DO

We are not a capital provider and will not pretend to be one. What a capital partner asks for is an independent technical opinion on the uplift: which wellbores, how much, and what the error bars are. That is inflow-performance and field-health work, and it is exactly what we do. An uplift number with error bars attached raises money that a flat claim does not.

SO-REG-003

ECONOMICS

Keep it, work it over, shut it in, or plug it — well by well

THE GAP

Marginal-well decisions get made on gut feel, or on one field-wide cost-per-barrel number, because doing it well by well is tedious. And almost every version of the sum leaves the plugging bill out — which quietly biases every answer toward "keep producing".

THE THING MOST MODELS GET BACKWARDS

Plugging cost is not *avoided* by producing. It is *deferred*. So a marginal well isn't competing against zero — it's competing against paying the P&A bill today, and deferral has real value. That is why a well can lose money every month and still be worth holding, and why "shut in" is sometimes the right answer when both producing and plugging are wrong.

WHAT YOU GET BACK

For each wellbore: month-one operating cash, the oil price at which it breaks even, the month it stops paying for itself, and all four courses of action priced on the same basis — every one of them ending with the well plugged, so the comparison isn't rigged.

WELL	MONTH 1 CASH	BREAKEVEN	WAY FORWARD
8 BOPD, 30 BWPD	+$9,412	$18.38/bbl	**Keep**
4 BOPD, 60 BWPD	+$3,154	$33.76/bbl	**Keep — 127 months left**
1.5 BOPD, 90 BWPD	−$1,014	$91.79/bbl	**Shut in**
0.6 BOPD, 120 BWPD	−$2,953	$260.03/bbl	**Shut in**

Illustrative cost deck — $65 oil, $45k plugging, 12.5% royalty, $0.75/bbl water — not any real operator's numbers. Note the bottom two: producing them destroys value, but shutting in still beats plugging by roughly $16,000 each, purely from deferring the plugging bill. That $16,000 is why so much of the country's stripper fleet is still standing.

OUR PROPOSAL · NOT YET ANYONE'S PRACTICE

Pool the spares before you pool anything else

Nobody asked us for this — we are proposing it, and we would rather be told it is wrong than have you assume other operators already agreed to it.

- **One canonical BOM.** A single rod-lift workover parts list with real sizes and grades, maintained as data, not as a text message.
- **A basin-level map of who holds what.** You do not need to buy a seat nipple that is sitting eleven miles away.
- **Shared reorder points.** The group carries one buffer instead of five partial ones — the same insurance for a fraction of the tied-up cash.
- **Group price discovery.** Individually you have no leverage on service pricing. Ten operators comparing the same line items do.

We would start it with the parts list and the map, because those are cheap and provable. Everything after that is a conversation among operators, not something an engineering firm should run.

WHAT WE DO AND DON'T DO

ASK	US?	WHY
Read my dynamometer cards and tell me what to change	Yes	Surface-to-downhole conversion, fillage, failure-mode diagnosis, setpoints
Rank my field so I know where to drive tomorrow	Yes	Field-health scoring across every well you send
Check whether my unit is overloaded	Yes	Torque balance and gearbox loading against the unit's rating
Independent technical review for a lender or partner	Yes	Inflow modelling and uplift estimates, with stated error bars
Tell me whether a well is worth keeping, working over, shutting in, or plugging	Yes	Well-by-well cash flow with the plugging bill carried in every case

ASK	US?	WHY
Find me money	No	We are engineers, not a placement agent
Tell me what my bonding obligation is, or how to satisfy it	No	That is legal and regulatory advice. We model the cost; your counsel handles the obligation
Identify corrosion from a photograph	Not alone	Photo ID is a judgement call — we bring the rate model behind it, not the verdict

STARTING COSTS YOU NOTHING BUT AN EXPORT

1. Send one well's card file — whatever your dyno or handheld exports. No install, no telemetry, no account.
2. You get back the downhole card, fillage stroke by stroke, the diagnosis with a confidence figure, and the parts of the card we do *not* trust.
3. If it is useful, send the field. If it is not, you have lost an email and we have learned something about your wells.

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